

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Quiz - 4
Duration: 25minutes

Semester: Summer 2025
Full Marks: 15

CSE 340: Computer Architecture

Name:	ID:	Section:
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1. Assume that a memory access takes 20 ns if it results in a cache hit, and 50 ns if it results in a cache miss. The current state of the direct-mapped cache is shown below:

Index	Valid	Tag	Data
000	1	100	Data[100000]
001	0		
010	0		
011	0		
100	1	100	Data[100100]
101	0		
110	1	011	Data[011110]
111	1	111	Data[111111]

The size of a block is 8 bits. Using the table below, calculate the total access time for the given memory sequence using a direct-mapped cache. Indicate hit/miss and access time for each entry. Write your final answers in the table below. [8]

Memory/Address	Hit/Miss	Access Time
36		
56		
26		
63		
30		

16		
29		
37		
Total Access Time		

a) Complete the cache table simulation in the given table above. [2]

b) Find the total access time. [1]

Q2: Suppose a cache has 16 blocks. Each block can store 48 bits of data.

a) What is the size of index bits? [1]

b) Using the mentioned cache system, for a memory address 1011100010, Find the index, tag and offset bits. [3]

Answer:

a)

b) Index bits =

Offset bits =

Tag bits =